

LEARNER WORKBOOK

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	ACHIEVED	NOT ACHIEVED	
KNOWLEDGE	☒	☐	<i>Well Done</i>
SKILLS	☐	☐	



Signature of learner



Signature of Assessor

COURTNEY ROSENBERG

Learner workbook



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This material has been compiled based on the **UNIT STANDARD 9012 INVESTIGATE LIFE AND WORK RELATED PROBLEMS USING DATA AND PROBABILITIES.**

The program has been accredited by SASSETA in the Republic of South Africa.

Material compiled by: Training Dynamics Africa

Learner workbook





HOW THIS LEARNER WORKBOOK IS COMPILED

This learner workbook has been compiled to provide credits towards the mathematical literacy requirement of the NQF at Level 3. The essential purposes of the mathematical literacy requirement are that, as the learner progresses with confidence through the levels, the learner will grow in a confident, insightful use of mathematics in the management of the needs of everyday living to become a self-managing person.

People credited with this unit standard are able to:

1. Pose questions, collect and organize data.
2. Represent and interpret data using various techniques to investigate real life and work problems.
3. Use random events to explore and apply probability concepts in simple life and work related situations.

You have to study the content and guidelines given in your learner reference guide to complete this workbook.

Remember that this completed workbook must be handed in as it will form part of your formative assessment and will play a role in awarding you the required 05 credits that are to be achieved upon completion of this module.

Write neatly and legibly and ensure that you do not simply copy the information contained in the learner reference guide into your workbook. Give your own opinion and ideas where applicable and write the answers in your own words to indicate your level of understanding and learning.

Good luck with the completion of this workbook.

LEARNING UNIT 1

Data

LEARNING OUTCOMES

A learning credited with this unit will be able to:

1. Explain the term sampling when producing data.
2. Explain what the term sampling methods.
3. Explain what surveys are and how to use it during the data producing phase.
4. Produce data within your own work environment by using the information in this learning guide.
5. Organize and analyse data that was produced within your work environment.
6. Display data and distributions as set out in this guide.
7. Represent data with respect to centre, spread and distribution.
8. Display data and distributions as set out in this guide.
9. Explain what measuring data with respect to validity, accuracy, and measurement scales is.
10. Draw a conclusion from your analysis.
11. Explain what index numbers are.
12. Explain what simple probability is.



LEARNER ACTIVITY 1

Consider the following table from Stats SA.

Population 16 years and above, in urban and non-urban areas, by province and gender (Stats

Province	Urban			Non-urban			Grand Total
	Female	Male	Unstated	Total	Female	Male	Total
Eastern Cape	879 370	769		1 648 950	1 264	856 390	3 769
Free State	718 041	579		1 274 035	007	330 422	347
Gauteng	2 615 535	555	11 028	5 248 206	203 533	237 852	1 807
Kwa-Zulu-Natal	1 555 122	993	17 000	3 171 798	74 220	830 317	989
Mpumalanga	366 049	2 621		754 074	1 404	470 510	5 560
North West	549 956	644		1 068 955	272	559 309	278
Northern Cape	250 891	1 599		451 817	563 378	70 791	5 406
Northern Prov- ince	174 450	675		322 551	579 434	1 039	386
Western Cape	1 349 576	388		2 580 090	38 031	747	1 787
		025			1 438	118 540	963
		518			552		2 207
		999			99 427		698
		200					560
		926					639
		148					2 810
		101					560
		1 230					2 798
		514					058
Total	8 458 992	8 033	28 028	16 520 476	5 664 854	4 513 878	26 708 918

- ♦ What does column 2 represent?

Total of females 16 years and older in the provinces



- ♦ What does column 5 represent?

Total of both males and females together in the provinces



- ♦ Which column represents the number of males living in non-urban areas?

Column 7



- ♦ If you wanted to find the total number of people, 16 years and older, living in a certain province, which column would you look in?

Column 10 - Grand Totals



- ♦ How many males, 16 years and older, live in urban areas in Kwa-Zulu-Natal?

1 555 993



Requirements achieved	YES				NO	
Comments in support of decision						

Achieved

LEARNER ACTIVITY 2

Every year millions of man days are lost because of people striking. A man day is one work day for one person. In other words, if 20 people strike for one day, 20 man days are lost. If 20 people strike for 2 days, 40 man days are lost. Look at this table of man days lost because of strikes:

Year	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
Man days lost (in millions)	9	1,5	3	4	3,5	4	3,5	1,5	1,5	0,5

- ♦ Draw a bar graph to represent the data in the table. Label the axes of your graph.

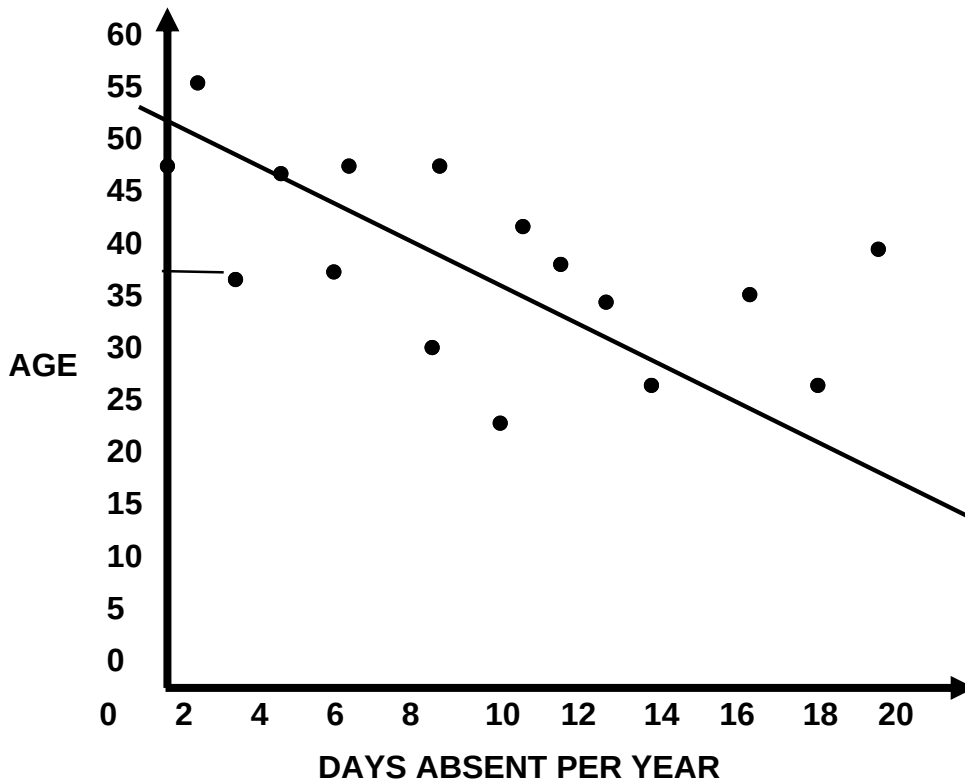


Requirements achieved	YES	☐	NO	☐
Comments in support of decision				

Achieved

LEARNER ACTIVITY 3

The Human Resources Manager of a company employing 200 workers conducts a survey to investigate whether there is a relationship between the age of workers and the number of days they are absent per year. He takes a sample of 15 files and records the age and the number of days absent of the workers in the sample. He then plots the results of the survey on the following scatter graph:


☐

- Do you think the department would decide that there **is** a correlation between the number of days absent and the age of workers? If yes, describe the correlation (positive or negative?) and explain.



Yes: Negative - The younger worker stays absent more often than the older workers

☐

- ♦ Use your ruler to draw a **line of best fit**.


☐

On the graph

- ♦ Using your line of best fit, estimate:

i) How many days a 35 year old worker can be expected to be absent in a year.

ii) The age of a worker who is likely to be absent for 3 days in a year.

i) Between 12 and 14 days about 13 days


☐

ii) Between 45 and 55 years about 50 years

Requirements achieved	YES		☐	NO
Comments in support of decision				

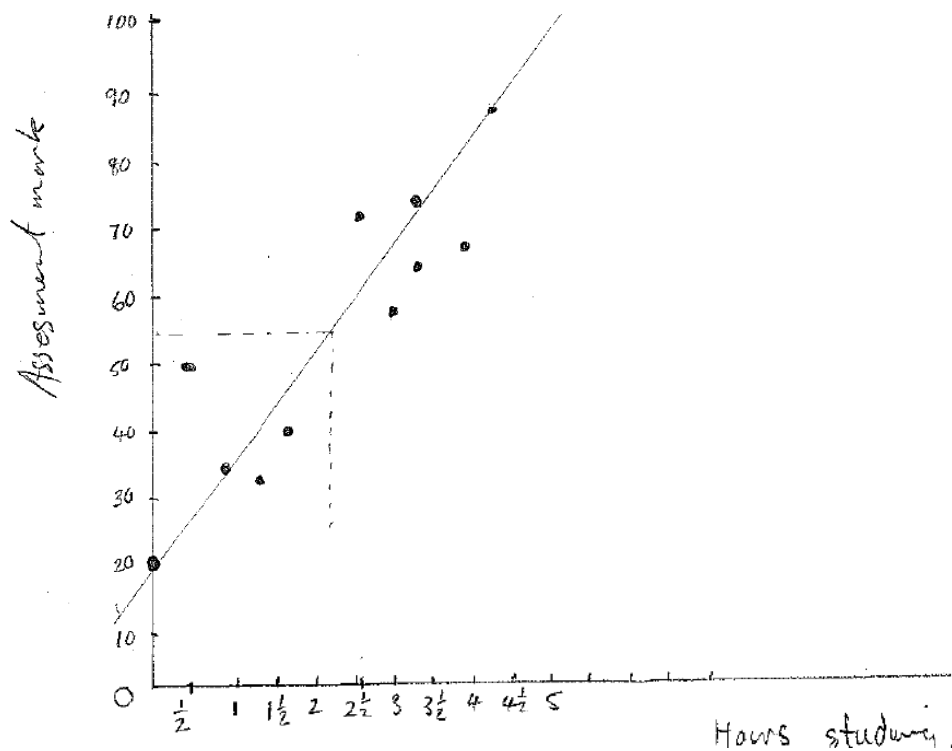
Achieved

LEARNER ACTIVITY 4

A Maths facilitator wanted to see whether there was a correlation between the numbers of hours that learners spent studying and the marks achieved for their summative assessments. She gathered the following information from 12 learners in her NQF 3 Maths class.

Hours spent studying	2	4	1	$1\frac{1}{2}$	0	2	$4\frac{1}{2}$	4	3	$3\frac{1}{2}$	5	$\frac{1}{2}$
Assessment mark	40	73	35	32	20	70	65	62	71	58	85	50

- ♦ Draw a scatter graph showing the information from the table.



- ♦ Is there a correlation between the numbers of hours that learners spent studying and the marks achieved for their summative assessments? If yes, describe it.


☐

- ♦ Use your ruler to draw a **line of best fit**.
- ♦ Use your line of best fit to predict:
- i) How long a learner who achieves 55% studied for.
- ii) What assessment mark someone who studies for $2\frac{1}{2}$ hours can expect.

i) $2\frac{1}{2}$ hours



ii) 55%

☐

Requirements achieved	YES		☐	NO
Comments in support of decision				

Achieved

LEARNER ACTIVITY 5

Three sets of data are given:

1. **Data Set 1:** 8 11 11 13 15 21 23
2. **Data Set 2:** 6 6 7 10 12 14 15 15
3. **Data Set 3:** 10 14 15 16 18 18 21 23 26

For each set of data find:

- a) The mode
- b) The median
- c) The mean (correct to one decimal place)
- d) The range
- e) The lower quartile
- f) The upper quartile
- g) The interquartile range

	Set	1	2	3
A	Mode	11	6 & 15	18
B	Median	13	11	18
C	Mean	14,7	10,6	17,9
D	Range	15	9	16
E	The lower quartile	11	14,5	6,5
F	The upper quartile	21	22	14,5
G	The interquartile range	10	7,5	8



Requirements achieved	YES	☒	☐	NO	☐
Comments in support of decision					

Achieved

LEARNER ACTIVITY 6

In a pack of cards, there are 52 cards. There are 4 suits: diamonds, hearts, spades and clubs. Each suit has a King, a Queen, a Jack and an Ace. It also has cards for 2, 3, 4, 5, 6, 7, 8, 9 and 10. If the pack has been shuffled so that the cards are in no particular order, and you select one card from the pack without looking, what is the probability of selecting:

- a) a Queen
- b) a red Queen
- c) a Jack of Hearts
- d) a Jack of Hearts or Diamonds
- e) any club
- f) a six or a seven
- g) a black six
- h) a picture card (Jack, Queen or King)

$$\begin{array}{lcl}
 \text{a)} & \frac{4}{52} & = \frac{1}{13} \\
 \text{b)} & \frac{2}{52} & = \frac{1}{26} \\
 \text{c)} & \frac{1}{52} & \\
 \text{d)} & \frac{2}{52} & = \frac{1}{26} \\
 \text{e)} & \frac{13}{52} & = \frac{1}{4} \\
 \text{f)} & \frac{8}{52} & = \frac{2}{13} \\
 \text{g)} & \frac{2}{52} & = \frac{1}{26} \\
 \text{h)} & \frac{12}{52} & = \frac{3}{13}
 \end{array}$$



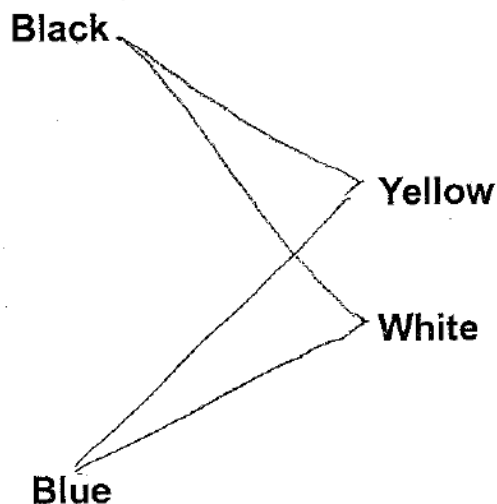
Requirements achieved	YES	☒	☐	NO	☐
Comments in support of decision					

Achieved

LEARNER ACTIVITY 7

A soccer team has two different coloured shorts and two different coloured shirts for their kit.

Complete the tree diagram below and then use it to fill in the table:



Possible outfit combinations	
<u>Shorts</u>	<u>Shirts</u>
Black	Yellow
Black	White
Blue	Yellow
Blue	White



- ♦ How many combinations are there? 4
- ♦ Calculate (in ratio form) the probability that the soccer team wear:
 - i) black shorts and white shirt
 - ii) blue shorts and yellow shirt
 - iii) green shorts
 - iv) yellow shirts
 - v) black shorts
 - vi) black shorts or white shirts
 - vii) blue shorts or yellow shirts



$$\begin{array}{lll}
 \text{i)} & \frac{1}{4} & 1:4 \\
 \text{ii)} & \frac{1}{4} & 1:4 \\
 \text{iii)} & 0 & \\
 \text{iv)} & \frac{1}{2} & 1:2 \\
 \text{v)} & \frac{1}{2} & 1:2 \\
 \text{vi)} & \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} & 1:4 \\
 \text{vii)} & \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} & 1:4
 \end{array}$$



Requirements achieved	YES		☐	NO	
Comments in support of decision					

Achieved

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

LEARNING OUTCOMES		1	2	3
1	Explain the term sampling when producing data.			✓
2	Explain what the term sampling methods.			✓
3	Explain what surveys are and how to use it during the data producing phase.			✓
4	Produce data within your own work environment by using the information in this learner guide.			✓
5	Organize and analyse data that was produced within your work environment.			✓
6	Display data and distributions as set out in this guide.			✓
7	Represent data with respect to centre, spread and distribution.			✓
8	Display data and distributions as set out in this guide.			✓
9	Explain what measuring data with respect to validity, accuracy, and measurement scales is.			✓
10	Draw a conclusion from your analysis			✓
11	Explain what index numbers are.			✓
12	Explain what simple probability is.			✓



Evidence Collector Signature

26.06.2021

Date



Security Officer Signature

21.06.2021

Date

Learner workbook



